

Freight Rate Prediction

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Approach and data

Predict posted_rate in USD from 48,000 labeled shipments dated January-October 2025. Final outputs cover 12,000 November-December shipments with no available labels. The selected main pipeline is CatBoost with an absolute-error objective and no quote-derived inputs. A separate reduced-input model supports the fixed-lane December chart.

Chronological model selection

Stage	Training / stopping window	Outer evaluation
Earlier fold	Jan-May / June; refit Jan-June	July and August
Later fold	Jan-July / August; refit Jan-Aug	September
Final refit	Jan-Oct; settings already frozen	Nov-Dec predictions

Boosting iterations are selected using the last month inside each training period, with patience 60. A fresh model is then fitted to the whole outer training window. July and August use the same January-June model. Configurations are ranked by equal-weight mean dollar MAE over July, August and September; mean RMSE breaks ties.

October had already been inspected during v1. It was excluded from v2 development and is not claimed as a new untouched holdout. Development scores select the pipeline and therefore are not unbiased final-test estimates. October enters only the final training refit after selection. Hidden-set accuracy is unavailable.

Cleaning and the negative-weight hypothesis

The labeled file contains 300 empty weights, 292 negative weights and 374 missing market-index values. Similar positive-weight and absolute-negative-weight distributions motivated testing sign correction, not assuming it. No clustering was performed; summaries group by the supplied equipment labels.

Four weight policies were compared for CatBoost: negatives missing or absolute, each with equipment-median imputation or native numeric NaN handling. Absolute negatives plus median imputation won this development run. Zero is treated as missing. Group medians and overall fallback are fitted only on training rows. Original-negative, original-empty and missing flags preserve provenance. Market index uses its training median and a missing flag.

The nearest weight-policy runner-up differs by only \$2.82 mean MAE. This one-seed result does not prove sign-entry mistakes or statistical significance. All target prices, including expensive shipments, remain unchanged.

Main model specification

Native categorical features: pickup, delivery and equipment. Numeric features: distance, weight, four coordinates, market index, weekday, month, day-of-month and provenance flags. Both quote_signal and quote_based_rate are excluded. IDs and target are excluded. CatBoost uses MAE loss, depth 6, learning rate 0.04, L2 regularization 5 and seed 42. Inner search is capped at 800 iterations. The no-quote pipeline used 778 trees in the earlier fold and 798 in the later fold; the final refit freezes 798. These values are close to the 800-tree search cap.

Measured Development Results

Selection history and quote ablation

The initial experiment compared Random Forest, LightGBM raw/log targets and CatBoost, then four weight policies for CatBoost. The selected v2 pipeline averaged \$122.39 monthly MAE versus \$149.46 for the RF rerun. The subsequent ablation specifically removed quote_signal and quote_based_rate together while retaining coordinates, market index and all other features.

Main pipeline	July MAE	Aug MAE	Sep MAE	Mean MAE
v2 with quote	120.68	134.44	112.04	122.39
v3 without quote - selected	113.64	113.43	118.49	115.18

Removing quote reduced equal-weight mean monthly MAE by **\$7.20 (5.89%)**. Mean monthly RMSE decreased from **\$624.73 to \$620.62**. July and August improved, while September MAE worsened by \$6.45. These results support the no-quote selection for this experiment; they do not establish leakage or guarantee better future performance.

Both variants were rerun with the same preprocessing, seed, hyperparameters and inner early-stopping rule. They selected their own iterations inside training: with quote 198/593, without quote 778/798. The with-quote run reproduced v2. Monthly metrics were independently recomputed from saved predictions. Mean monthly RMSE is not pooled-row RMSE.

Limits of the additional experiment

The ablation was added after reviewing v2 and reused July-September development labels. Choosing again on these months increases selection bias; neither these scores nor October constitute a new independent test. This was one seed, with an 800-tree budget and patience 60. The near-cap iterations are a constraint; no 3,000-tree extension was run. XGBoost and LSTM were not tested.

Reduced-input December model - unchanged

Even without quote, the main model requires coordinates and market index, which the chart input lacks. The separately evaluated reduced CatBoost model therefore remains necessary. Its inputs are pickup, delivery, equipment, distance, weight, calendar features and weight flags.

Reduced pipeline	July MAE	Aug MAE	Sep MAE	Mean MAE
Basic Random Forest	172.89	223.00	138.50	178.13
CatBoost / absolute + median	136.74	193.85	108.12	146.24

Reduced CatBoost mean monthly RMSE is \$625.62 versus \$639.45 for RF. Its earlier inner fold selected 800 trees (the cap); the later fold selected 790. The final model retains 790 trees. Refitting on all labeled rows reproduced the v2 December predictions exactly. No settings were changed for this model.

Evidence and provenance

The package preserves original v2 evidence and the separate quote-ablation protocol, metrics and row-level predictions. The current final_run.json records v3 features, settings, versions and hashes. Research references are listed in SOURCES.md. External repository scores were not used as measured results. Source data are not bundled.

December Chart and Delivery

Fixed lane: Lexington to Fort Wayne

360 miles | Dry Van | 32,000 lb | December 1-31, 2025. Only the date changes across the 31 input rows. Predictions use the separately selected reduced CatBoost model.

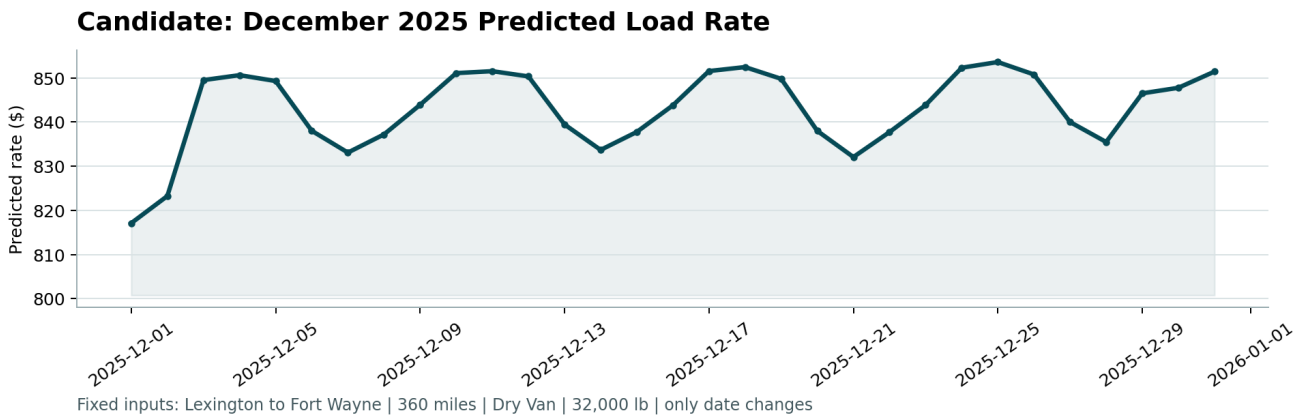


Figure 1. Organizer-generated chart from the unchanged December predictions. Rates range from \$817.10 to \$853.58. The vertical axis is truncated and emphasizes small daily changes.

Interpretation and practical limits

These are predictions, not observed December prices. With no labeled November-December history, trees cannot extrapolate a reliable unseen-month trend. Daily variation reflects fitted weekday/day-of-month effects; no holiday-specific model or uncertainty interval is included. Global reduced-model scores do not establish accuracy on this single lane.

The final main model no longer depends on quote fields. The ablation measures predictive usefulness under this protocol, not the provenance of quote_signal; removing it does not establish that it was leaked. Rare expensive shipments remain difficult and were retained. Further evaluation should use genuinely new labeled periods; repeated tuning on October would not produce a fresh test.

Verified outputs and reproduction

The unchanged organizer scorer validated exactly 12,000 unique expected IDs with positive finite rates and 31 December predictions with the required columns, dates and fixed inputs. Additional checks confirmed template ID order and preservation of the original December input fields. The scorer validates files and draws the chart; Spotter calculates hidden-set accuracy after submission.

The package includes code, pinned direct dependencies, outputs, development evidence and a Loom script. Python 3.12.14 on Linux used four CPU threads. README.md gives commands: train.py refits the frozen models into rerun/ by default; quote_ablation/run_ablation.py reproduces the quote comparison. Existing prediction files require explicit overwrite. Models are not serialized.

Upload the package contents to an accessible GitHub repository, record the 2-3 minute walkthrough, and submit the required links and artifacts through the assessment channel. Publication and recording remain separate actions.